Worksheet 7 · Algebraic Fractions IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Simplify algebraic fractions by factorising • Add, subtract, multiply and divide algebraic fractions • Solve equations involving algebraic fractions		
Grade: / 50	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Algebraic Fractions*

Question 1 (3 marks)

Express $\frac{5}{3} - \frac{x+2}{2x}$ as a single fraction in its simplest terms.

Question 2 (3 marks)

(a) Solve $\frac{4-3x}{5} - \frac{3x-5}{2} = -3$

Show clear algebraic working.

$x =$ .....

(3)

Question 3 (3 marks)

Simplify fully $\frac{x^2-7x+12}{4x-x^2}$

(3)

Question 4 (3 marks)

Simplify fully $\frac{6x^3+13x^2-5x}{4x^2-25}$

Question 5 (3 marks)

Express $\frac{1}{9x^2-25} - \frac{1}{6x+10}$ as a single fraction in its simplest form.

(3)

Question 6 (4 marks)

Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

Question 7 (4 marks)

Express $\frac{4x^2-25}{5x^2+2x-7} \times \left(\frac{2}{x-3} - \frac{3}{2x-5}\right)$ as a single fraction in its simplest form.

Question 8 (3 marks)

Express

$$\frac{3}{x} + \frac{x+2}{2x} + \frac{1}{4}$$

as a single fraction in its simplest form.

(3)

Question 9 **(2 marks)**

Write as a single fraction $\frac{2}{3x} + \frac{4}{5x} - \frac{9}{10x}$

Give your answer in its simplest form.

.....

(2)

Question 10 **(3 marks)**

Express $\frac{4}{x-2} - \frac{3}{x+1}$ as a single fraction.

Give your answer in its simplest form.

.....

(3)

Question 11 **(3 marks)**

(a) Solve $\frac{9a-7}{5} - \frac{3a-7}{4} = 4.55$

Show clear algebraic working.

$a =$ .....

(3)

Question 12 **(4 marks)**

(a) Solve $\frac{4x+5}{3} - \frac{3-2x}{2} = 13$

Show clear algebraic working.

$x =$ .....

(4)

Question 13 **(3 marks)**

$a = \frac{14}{3x-7}$ $x = \frac{7}{4y-3}$

Express a in the form $\frac{py+q}{ry+s}$ where p, q, r and s are integers.

Give your answer in its simplest form.

$a =$ .....

Question 14 **(5 marks)**

Solve the equation

$$\frac{5}{x+2} + \frac{3}{x^2+2x} = 2$$

Show clear algebraic working.

Question 15 **(4 marks)**

Given that $x = \frac{5}{9y+5}$ and that $y = \frac{5}{5a-2}$

find an expression for x in terms of a.

Give your expression as a single fraction in its simplest form.